

Exp no:08 Use Steg-Expose to Detect Hidden Data in Images

Date:

Description

StegExpose is a specialized tool used for steganography analysis. It works by evaluating the statistical properties of an image to estimate the probability of hidden data being embedded within it. This process helps forensic investigators detect steganography techniques like Least Significant Bit (LSB) embedding.

Prerequisites

- Java Runtime Environment (JRE): StegExpose is a Java-based application and requires JRE to run.
- StegExpose Tool: Download the latest .jar file from the official StegExpose GitHub repository.

Step-by-Step Process

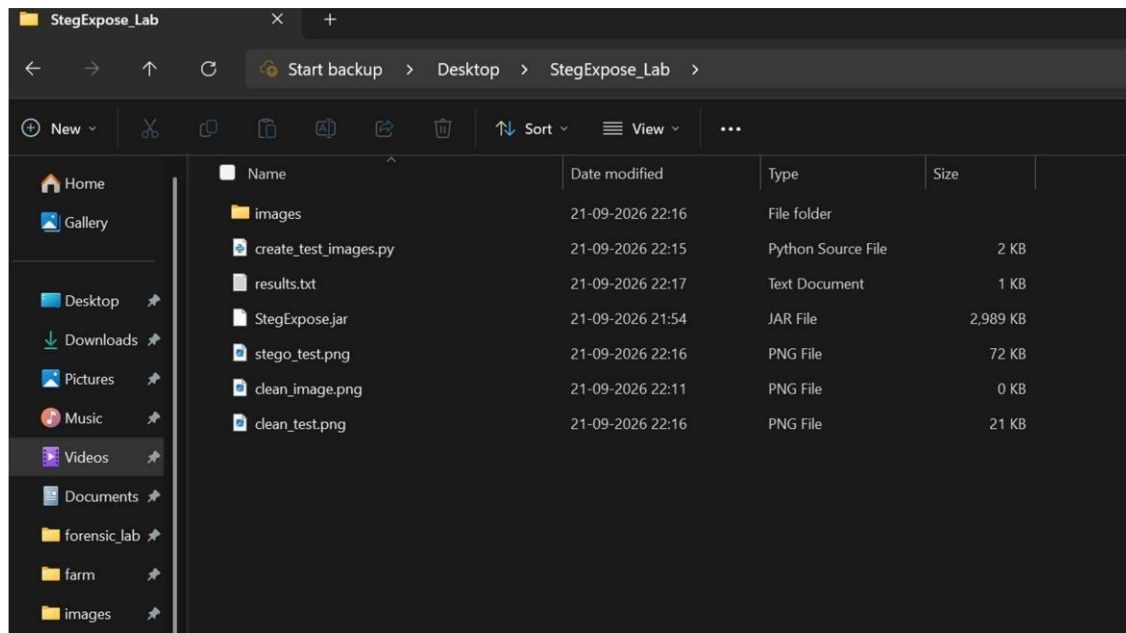
1. Download and Set Up StegExpose

1. Download the tool: Obtain the StegExpose.jar file from the GitHub repository.
2. Install Java: Ensure JRE is installed on your machine.
3. Prepare environment: Place the StegExpose.jar file in a dedicated working folder.



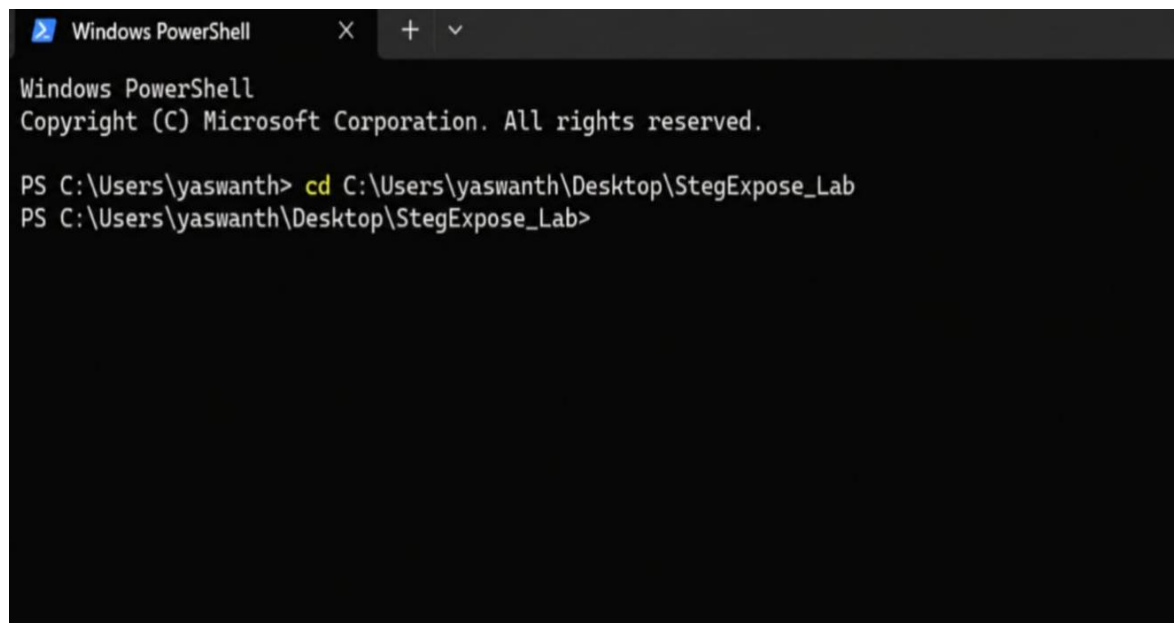
2. Select Images for Analysis

- Collect the images you suspect might contain hidden data.
- StegExpose supports common image formats such as .png, .jpg, and .bmp.



3. Open Command Line or Terminal

- Navigate to the folder where the StegExpose.jar file is located using your Command Prompt (Windows) or Terminal (Linux/macOS).



4. Run StegExpose on an Image

- Use the following command structure to analyze a single image for hidden data:

Command Structure:

```
java -jar StegExpose.jar <image_file_path>
```

Example:

```
java -jar StegExpose.jar test_image.png
```

5. Analyze the Output

- StegExpose calculates a "suspect" score ranging from 0 to 1. The higher the score, the more likely steganography is present.

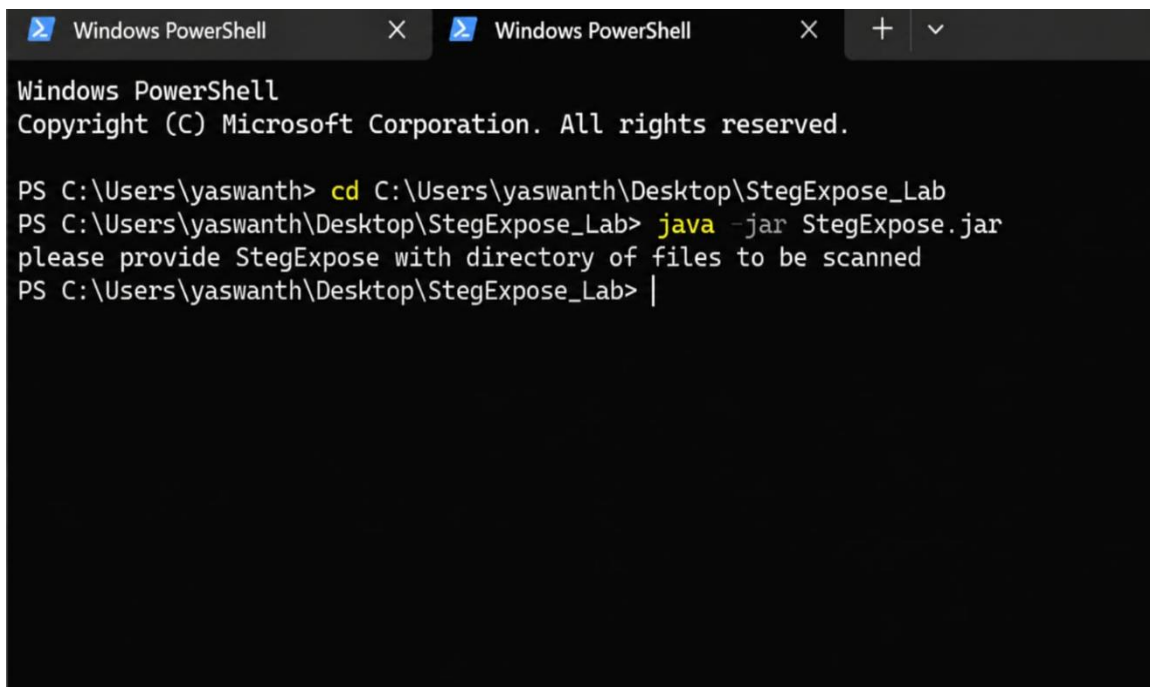
Score Range	Interpretation (Suggested Thresholds)
Less than 0.2	Image is considered clean (no hidden data detected).
0.2 - 0.3	Possibly some hidden data is present.
Above 0.3	Likely that steganography is present.

- Example Output Analysis: java -jar StegExpose.jar
suspect_image.png

Analyzing suspect_image.png...

Result: 0.4

Steganography likely present

A screenshot of a Windows PowerShell terminal window. The window has two tabs, both labeled 'Windows PowerShell'. The terminal text shows the user navigating to the directory 'C:\Users\yaswanth\Desktop\StegExpose_Lab' and running the command 'java -jar StegExpose.jar'. The program responds with 'please provide StegExpose with directory of files to be scanned'. The prompt is currently at 'PS C:\Users\yaswanth\Desktop\StegExpose_Lab> |'.

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\yaswanth> cd C:\Users\yaswanth\Desktop\StegExpose_Lab
PS C:\Users\yaswanth\Desktop\StegExpose_Lab> java -jar StegExpose.jar
please provide StegExpose with directory of files to be scanned
PS C:\Users\yaswanth\Desktop\StegExpose_Lab> |
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\yaswanth> cd C:\Users\yaswanth\Desktop\StegExpose_Lab
PS C:\Users\yaswanth\Desktop\StegExpose_Lab> java -jar StegExpose.jar
please provide StegExpose with directory of files to be scanned
PS C:\Users\yaswanth\Desktop\StegExpose_Lab> java -jar StegExpose.jar images
stego_test.png is suspicious. Approximate amount of hidden data is 5325 bytes.
PS C:\Users\yaswanth\Desktop\StegExpose_Lab> |
```

6. Batch Analysis (Multiple Images)

- To check multiple images at once, specify the folder path containing the images:\

Command Structure:

```
java -jar StegExpose.jar <folder_path>
```

7. Advanced Options (Optional)

- To view additional parameters, such as options for adjusting detection sensitivity or output verbosity, use the --help flag:

Command:

```
java -jar StegExpose.jar --help
```

8. Review the Results

- Review the scores generated for each image and use the threshold values to determine which images require further forensic investigation to extract the suspected hidden data.

Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

The hidden data within the image was successfully detected using StegExpose. The analysis revealed that images with a suspect score above the threshold likely contain embedded steganographic content, confirming the tool's effectiveness in steganography detection.